ME-205 FLUID MECHANICS										
L	**T**	**P**	Credit	Area		**CWS**	**PRS**	**MTE**	**ETE**	**PRE**
3	0	2	4	DCC		15	25	20	40	-

Course Outcomes: After completing the course, student will be able to

CO1	Understand the fluid properties, flow types and concept of hydrostatic forces, concept of flow lines, mass conservation principle, flow net sand vortex flow basics.
CO2	Understand Reynolds's transport theorem, Euler's equation, Bernoulli's equation and its application and also Buckingham's pi Theorem.
CO3	Describe the concept and analysis of laminar flow in various flow problems.
CO4	Describe the concept and analysis of turbulent flow in various flow problems.
CO5	Analyze the concept of Boundary layer flow theory and about flow around immersed body.
CO6	Apply the basic concept of hydraulic turbines and pumps.

Syllabus		**Contact Hour**
Unit-1	**Introduction:** Types of fluid and flow, continuum, fluid properties. **Fluid Statics:** Pressure variation in a static fluid; hydrostatic manometry; forces on planes and curved surfaces, stability of submerged and floating bodies. **Fluid kinematics**: General description of fluid motion, steady flow, uniform flow; stream, streak and path lines; Lagrangian and Eulerian approach; Continuity equation, particle acceleration; rotational and irrotational flow; stream function; velocity potential function, flow nets; circulation; simple flows; source, sink, vortex, doublet, free and forced vortex.	**10**
Unit-2	**Fluid Dynamics:** Concept of system and control volume; Reynolds's transport theorem, Euler's equation, Bernoulli's equation, Navier Stokes equation; Flow measurement-Venturi meter, Orifice meter, Pitot-tube, flow meters, notches. **Dimensional analysis:** Buckingham's –pi Theorem. Non-dimensional parameters, similarity and its application to fluid problems.	**7**
Unit-3	**Viscous flow:** Laminar flow between parallel surfaces and through the circular pipes, Momentum and Kinetic energy correction factors; power absorbed in viscous resistance, film lubrication	**7**
Unit-4	**Turbulent flow:** Transition from laminar to turbulent flow, turbulence and turbulence intensity, turbulence modelling, Prandtl mixing length hypothesis; flow losses in pipes-major and minor losses, pipes in series And parallel, hydraulically smooth and rough pipes, friction factor charts.	**7**
Unit-5	**Laminar and Turbulent Boundary Layer flows**: Boundary layer concept, boundary layer thickness, displacement, momentum and energy thickness. Momentum integral equation; dragon flat plate. Boundary separation. Flow around immersed bodies- drag and lift.	**7**
Unit-6	**Hydraulic Turbines and Pumps**: Impact of Jet on Flat, Curved & Moving Plates – Hydraulic turbines; Definition and Classification, Hydraulic pumps; Definition and Classification, Euler's turbo-machine equation.	**4**
	Total	**42**

Reference Book:	
1	Introduction to Fluid Mechanics and Fluid Machines, Som. S. K & Biswas, G, Tata McGraw-Hills Publishing Company Limited, ISBN-0074633716,2003.
2	Fluid Mechanics, Cengel & Cimbala, Tata McGraw-Hills. Third Edition, ISBN-0070700346,2015.
3	Fluid Mechanics, White F. M, Tata McGraw-Hill Publishing Company Limited. Seventh Edition, ISBN-0071333126,2011.
4	Engineering Fluid Mechanics, Prof K. L. Kumar, S Chand, ISBN 9788121901000
5	Introduction to Fluid Mechanics, Fox and Pritchard, Seventh Edition. Wiley India, ISBN-9788126523177.
6	Hydraulic Machines-by K Subramanya, 2014, McGraw hill, ISBN-10:1-25-900684-0

CO-PO/PSO Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	1	1	2	1	1	1	1	2	3	2	1
CO2	3	2	2	1	2	2	2	1	1	2	2	2	3	2	1
CO3	3	3	3	2	1	1	2	1	2	2	1	2	2	3	2
CO4	3	3	3	2	1	1	2	1	1	1	1	2	3	2	1
CO5	3	3	2	2	1	1	2	1	1	1	1	2	3	2	1
CO6	3	2	2	2	1	1	2	1	2	1	1	2	2	2	1